2-Step Hierarchal Model

step 1) collect lots of source observations under a very wide prior $p(\theta)$.

$\Rightarrow$ posterior for each source $p(\vec{\theta}_i \mid d_i)$

step 2)

$$p(\phi / d) \propto \int \prod_i \frac{p(\theta_i \mid d_i)}{p(\theta)} \, p(\theta \mid \phi) \times p(\phi)$$

Reweighted posterior

In LVK, we have $\mathcal{O}(300)$ sources, so we have a good idea of population! $p(\phi)$.

$\Rightarrow$ We analyze every event as if we did not know the population, including masses that don't exist in the catalogs!

① Analyze each event independently

② Reweight w/ conditional prior $p(\theta \mid \phi)$

It's a choice! Sacrificing knowledge on individual events for knowledge about the population.

$\Rightarrow$ There are other ways of doing joint hierarchical inference, where we learn the pop. and the sources at the same time.

(This is hard! Active research area)

Only way that could work for LISA

Stochastic Sampling Algorithms

prior $p(\theta)$

likelihood $p(d|\theta)$ $\Longrightarrow$ posterior $p(\theta|d)$

evidence $p(d)$

Any method that can accomplish this!

We're not just finding the best fit, but the whole distribution of good-fitting parameters. (posterior $p(\theta|d)$).